

Nishant Ajit

7619525006 — [Email](#) — [Github](#) — [Portfolio](#)

EDUCATION AND SCHOLASTIC ACHIEVEMENTS

VIT, Vellore

2023–2027

Bachelor of Technology in Computer Science (IoT) – 8.62 CGPA

Gear Innovative International School

2023

12th Grade (CBSE) – 95.2%

EXPERIENCE

Health Vectors

Summer 2026

AI Product Engineer Intern

- Conducted requirement gathering with the CEO and CTO of a multi-geography B2B health-tech company, translating ambiguous pain points around a growing pipeline (130+ leads and clients across India, GCC, and Malaysia) into a fully specified product.
- Designed and shipped an AI-powered sales dashboard replacing a legacy Excel lead sheet, owning architecture, data schema, API design (15+ endpoints), and UI/UX across 4 screens; decisions validated with CTO before build
- Integrated OpenAI Whisper (STT) and an LLM pipeline enabling the CEO to dictate client interactions by voice; system automatically extracts tasks and summarises notes, eliminating manual data entry
- Implemented Microsoft Azure AD authentication (MSAL) with an email allowlist, restricting access to authorised company users only
- Containerised the full stack (FastAPI backend + Next.js frontend) using Docker. Managed secrets via Infisical vault, keeping all credentials out of source code
- System is actively in daily use by the CEO and his team across phone and laptop

PROJECTS

[NotesAssistant](#) — **RAG-based Study Material Q&A System**

- Engineered a Retrieval-Augmented Generation system for semantic search over lecture materials using LangChain and FAISS vector indexing
- Implemented batch processing for document embedding to reduce memory usage from >4GB to <1GB for large PDF files
- Leveraged BGE embeddings with recursive text splitting for context-preserving document chunking

[Spotify Helper Agent](#) — **Natural Language Music Control**

- Built a ReAct agent using LangChain and Gemini that translates natural language into Spotify Web API actions
- Integrated OAuth 2.0 authentication flow for secure user authorization and token management
- Extended agent capabilities with Last.fm and SerpAPI tools for music discovery and recommendations

[EdgeAQI](#) — **Air Quality Monitoring and Mapping with Edge-Fog Architecture**

- Developed a three-layer edge-fog system for real-time location-aware air quality heatmap generation using ESP32, MQ-135, and DHT11 sensors
- Offloaded GPS processing to a smartphone fog node, eliminating dedicated GPS hardware cost on the sensor node while also reducing power consumption and improving location accuracy
- Mapped GPS-tagged air quality readings as spatial hotspots to identify pollution concentration zones across a given area

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Java, SQL
- **AI/ML Stack:** LangChain, RAG, FAISS, BGE Embeddings
- **Backend Development:** FastAPI, MongoDB, JWT Authentication, REST APIs, OAuth 2.0
- **Developer Tools:** Git, Docker, Bash, Azure MSAL